

HUMAN PHYSIOLOGY - I

Section 2: Short Notes (Answer any six)

Organised exam notes with core points, key values and source diagrams

How to use: For a 5-mark short note, write the definition first, then 4-6 core points, add one table/diagram if possible, and finish with one applied/clinical point.

Quick index

No.	Topic	Diagram / visual added
1	Oxygen-haemoglobin dissociation curve	O ₂ -Hb curve with left/right shift
2	Structure of cell membrane	Fluid mosaic membrane + lipid layer
3	Functions of WBC	Clean functional table
4	Acquired immune system	Development of acquired immunity
5	Functions of saliva	Composition of saliva
6	Cardiac cycle	Events of cardiac cycle
7	Normal ECG	Waves of normal ECG

2(a). Oxygen-Haemoglobin Dissociation Curve

Definition

The oxygen-haemoglobin dissociation curve shows the relationship between partial pressure of oxygen (PO₂) and percentage saturation of haemoglobin with oxygen. It explains the affinity of haemoglobin for oxygen.

Normal curve

- **Shape:** S-shaped or sigmoid, because oxygenation of one haem group increases the affinity of remaining haem groups.
- **In lungs:** High PO₂ favours loading of oxygen on haemoglobin.
- **In tissues:** Low PO₂ favours unloading of oxygen from haemoglobin.

PO ₂	Hb saturation	Importance
25-27 mm Hg	50%	Called P ₅₀ ; indicates affinity of haemoglobin for oxygen.
40 mm Hg	About 75%	Typical venous blood level; oxygen is released to tissues.
100 mm Hg	About 95%	Typical arterial blood level; haemoglobin is almost fully saturated.

Shift of the curve

Right shift	Left shift
Means decreased affinity of Hb for O ₂ ; oxygen is released easily to tissues.	Means increased affinity of Hb for O ₂ ; oxygen is held more tightly.
Causes: increased CO ₂ , increased H ⁺ ions, decreased pH, increased temperature, increased 2,3-DPG.	Causes: decreased CO ₂ , decreased H ⁺ ions, increased pH, decreased temperature, fetal haemoglobin.
Useful during exercise and in metabolically active tissues.	Useful in fetus because fetal Hb can take oxygen from maternal blood.

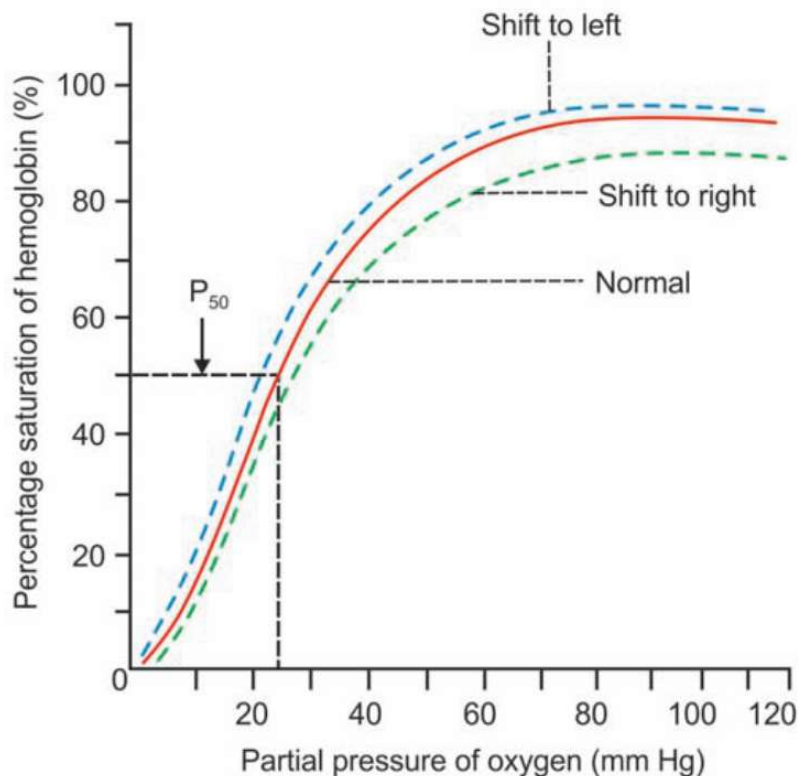


FIGURE 125.1: Oxygen-hemoglobin dissociation curve

Source diagram: Oxygen-haemoglobin dissociation curve with left and right shifts (Sembulingam, Fig. 125.1).

Applied note

Right shift helps active tissues receive more oxygen. Left shift improves oxygen uptake but reduces oxygen delivery to tissues.

High-yield exam tips

- Bohr effect: increased CO_2 and H^+ shift the curve to the right and promote oxygen unloading.
- P_{50} is the PO_2 at which haemoglobin is 50% saturated; increased P_{50} means decreased affinity.
- Carbon monoxide shifts the curve left but reduces oxygen-carrying capacity, causing tissue hypoxia.
- 2,3-DPG increases in chronic hypoxia and high altitude, helping tissues receive more oxygen.

2(b). Structure of Cell Membrane

Definition

The cell membrane, also called plasma membrane or plasmalemma, is a protective semipermeable membrane separating intracellular fluid from extracellular fluid.

Composition

Component	Approximate percentage	Main role
Proteins	55%	Channels, carriers, pumps, receptors, enzymes, antigens and adhesion molecules.
Lipids	40%	Phospholipid bilayer forms the basic barrier; cholesterol maintains stability.
Carbohydrates	5%	Form glycocalyx; help in cell recognition, cell adhesion and receptor function.

Fluid mosaic model

According to the fluid mosaic model, the membrane is a fluid lipid bilayer in which proteins are embedded or floating. This model explains membrane flexibility, selective permeability and movement of molecules within the membrane.

Lipid bilayer

- **Phospholipids:** Have hydrophilic heads facing ECF/ICF and hydrophobic tails facing each other in the centre.
- **Cholesterol:** Packed between phospholipid molecules; provides structural integrity and controls fluidity.
- **Permeability:** Fat-soluble substances like O₂, CO₂ and alcohol pass easily; water-soluble substances need channels or carriers.

Membrane proteins

Type	Location	Examples / functions
Integral proteins	Pass through entire membrane thickness	Channels, carriers, pumps, receptors, cell adhesion proteins, antigens.
Peripheral proteins	Partly attached to inner or outer surface	Cytoskeletal proteins, some enzymes and transport proteins.

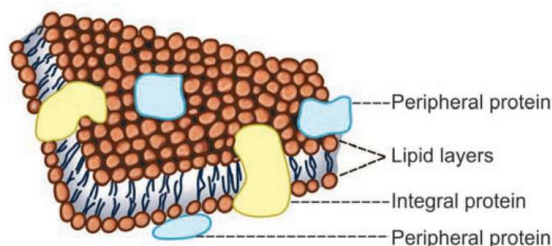


FIGURE 1.2: Diagram of the cell membrane

Source: Cell membrane (Fig. 1.2).

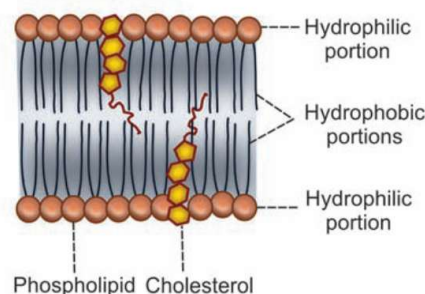


FIGURE 1.3: Lipids of the cell membrane

Source: Lipid layer (Fig. 1.3).

Functions of cell membrane

- Protects cytoplasm and organelles.
- Acts as a selectively permeable barrier.
- Allows absorption of nutrients and excretion of metabolic wastes.
- Permits exchange of gases between cell and extracellular fluid.
- Maintains the shape and size of the cell.

Extra important points

- Integral proteins commonly act as ion channels, carrier proteins, pumps and receptors.
- Peripheral proteins mainly support the cytoskeleton and some enzymatic activities.
- Glycocalyx is important for cell recognition, cell adhesion and hormone receptor activity.
- A good labelled diagram should show phospholipid bilayer, hydrophilic heads, hydrophobic tails, cholesterol, integral protein and peripheral protein.

2(c). Functions of White Blood Cells (WBC)

Core idea

White blood cells protect the body against invading microorganisms, foreign substances and abnormal cells. They form an essential part of the body defence mechanism.

General properties useful for defence

Property	Meaning
Diapedesis	WBCs pass through capillary walls into tissues.

Property	Meaning
Amoeboid movement	WBCs move by changing their shape.
Chemotaxis	WBCs move towards chemicals released at injured or infected sites.
Phagocytosis	WBCs engulf and destroy bacteria, dead cells and foreign particles.

Cell-wise functions

WBC type	Major functions
Neutrophils	First line of defence in acute bacterial infection; phagocytose and destroy microorganisms using enzymes and antimicrobial substances.
Monocytes / macrophages	Phagocytosis, removal of dead cells, antigen presentation and cytokine secretion.
Eosinophils	Defence against parasites; control allergic reactions; destroy worms and some antigen-antibody complexes.
Basophils	Release histamine, heparin, serotonin and other mediators; important in allergy and inflammation.
Lymphocytes	Responsible for immunity: B cells produce antibodies, T cells provide cellular immunity, NK cells destroy abnormal cells.

Phagocytosis sequence

Chemotaxis -> Diapedesis -> Opsonization -> Phagocytosis -> Killing/digestion of the foreign body.

Applied note

Increase in neutrophils usually suggests acute bacterial infection, while increase in eosinophils is commonly associated with allergy or parasitic infestation.

Extra values and clinical tips

Point	Remember
Normal total leukocyte count	About 4,000-11,000 cells/mm ³ of blood.
Neutrophils	Most abundant WBC; rise mainly in acute bacterial infection.
Lymphocytes	Important in viral infections and long-term immunity.
Eosinophils	Rise in allergy, asthma and parasitic infestation.
Monocytes	Become macrophages in tissues; important in chronic inflammation.
Basophils/mast cells	Release histamine and heparin; important in allergy and inflammation.

2(d). Acquired Immune System

Definition

Acquired immunity is the specific immunity that develops after exposure to a particular antigen. It has specificity and memory, so the second response to the same antigen is faster and stronger.

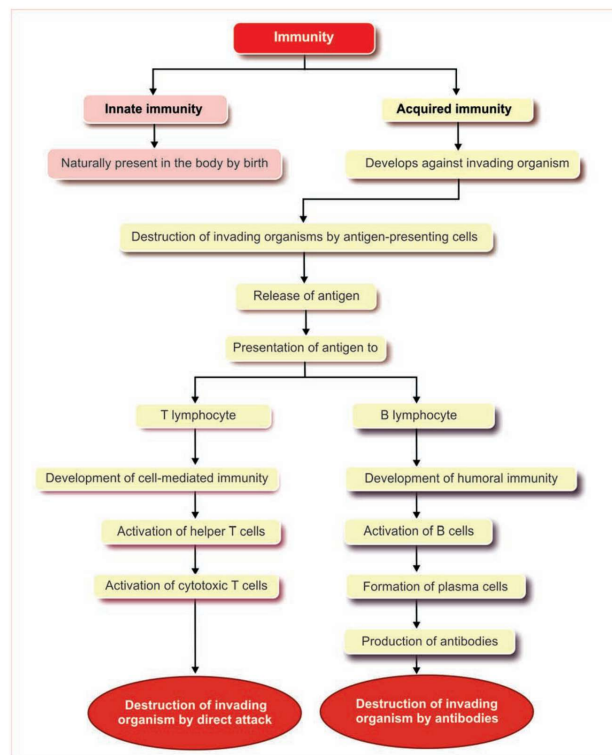


FIGURE 17.1: Schematic diagram showing development of immunity

Source diagram: Development of innate and acquired immunity (Sembulingam, Fig. 17.1).

Main cells

Cell	Role
T lymphocytes	Responsible for cell-mediated immunity; processed in thymus.
B lymphocytes	Responsible for humoral immunity; form plasma cells and antibodies.
Antigen-presenting cells	Process antigen and present it to lymphocytes to start acquired immune response.

Types of acquired immunity

Type	Mechanism	Example	Duration
Active immunity	Body produces its own antibodies or sensitized lymphocytes.	Natural infection or vaccination.	Slow onset but long-lasting.
Passive immunity	Ready-made antibodies are received from outside.	Maternal antibodies or antiserum.	Immediate but short-lasting.

Cell-mediated and humoral immunity

- **Cell-mediated immunity:** T cells directly attack infected cells, tumour cells and foreign cells; helper T cells coordinate immune response.
- **Humoral immunity:** B cells become plasma cells and produce antibodies that neutralize or destroy antigens.
- **Memory cells:** Remain after first exposure and produce a rapid secondary response.

Exam line

Acquired immunity is antigen-specific, has memory and is mediated mainly by T lymphocytes and B lymphocytes.

Primary and secondary immune response

Feature	Primary response	Secondary response
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Feature	Primary response	Secondary response
Exposure	First exposure to antigen.	Second/subsequent exposure to same antigen.
Speed	Slow onset due to lag period.	Rapid onset due to memory cells.
Antibody level	Lower antibody titre.	Higher antibody titre.
Main antibody	IgM appears first.	IgG predominates.
Importance	Basis of initial immune activation.	Basis of booster dose response.

2(e). Functions of Saliva

Introduction

Saliva is an essential digestive juice secreted by salivary glands. Mixed saliva contains about 99.5% water and 0.5% solids, including enzymes, mucus, ions and protective substances.

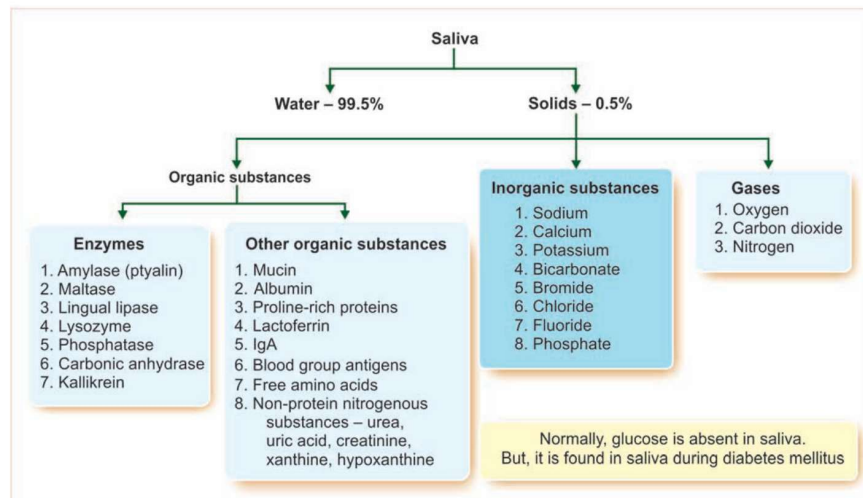


FIGURE 37.3: Composition of saliva

Source diagram: Composition of saliva (Sembulingam, Fig. 37.3).

Functions

Function	Explanation
Preparation of food for swallowing	Moistens food and mucus lubricates the bolus, making chewing and swallowing easier.
Taste sensation	Dissolves food substances so they can stimulate taste buds.
Digestive function	Salivary amylase starts starch digestion; lingual lipase helps lipid digestion, especially in infants.
Cleansing action	Washes oral cavity and removes food debris.
Protective action	Lysozyme, lactoferrin, IgA and other factors inhibit bacterial growth.
Speech	Moistens mouth, tongue and lips, helping articulation.
Excretory function	Certain substances like iodide, mercury, lead, viruses and some drugs may be excreted in saliva.
Water balance	Reduced salivation causes dryness of mouth and stimulates thirst.

Applied note

Absence or reduction of saliva causes dryness of mouth, difficulty in swallowing, loss of taste, dental caries and increased oral infection.

Saliva quick facts

- Daily salivary secretion is roughly 1-1.5 L/day in adults.
- Parotid saliva is watery and enzyme-rich; submandibular and sublingual secretions contain more mucus.
- Parasympathetic stimulation produces abundant watery saliva; sympathetic stimulation produces thicker saliva.
- Salivary amylase acts best near neutral pH and becomes inactive in highly acidic gastric contents.

2(f). Cardiac Cycle

Definition

Cardiac cycle is the sequence of coordinated events occurring in the heart during one heartbeat. It consists of systole and diastole of atria and ventricles.

Duration

At a normal heart rate of 72 beats/minute, one cardiac cycle lasts about 0.8 second.

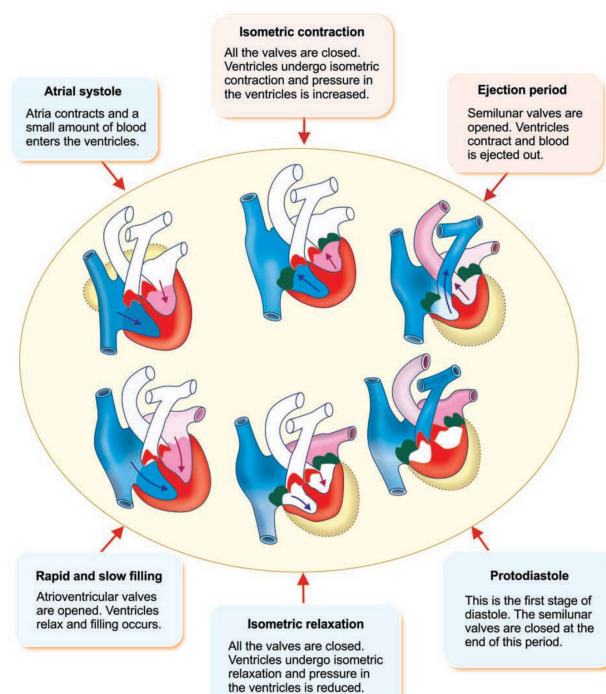


FIGURE 91.2: Events of cardiac cycle

Source diagram: Events of cardiac cycle (Sembulingam, Fig. 91.2).

Event	Duration	Meaning
Atrial systole	0.1 sec	Atria contract and complete ventricular filling.
Atrial diastole	0.7 sec	Atria relax and receive venous blood.
Ventricular systole	0.27 sec	Ventricles contract and eject blood.
Ventricular diastole	0.53 sec	Ventricles relax and fill with blood.

Phases of ventricular cycle

Phase	Main event	Valve status / heart sound
Isometric contraction	Ventricular pressure rises without change in volume.	All valves closed; AV valve closure causes first heart sound.
Ejection period	Blood is ejected into aorta and pulmonary artery.	Semilunar valves open.

Phase	Main event	Valve status / heart sound
Protodiastole	Early ventricular relaxation begins.	Semilunar valves start closing.
Isometric relaxation	Ventricular pressure falls without change in volume.	All valves closed; semilunar valve closure causes second heart sound.
Rapid filling	Blood rapidly enters ventricles.	AV valves open.
Slow filling / diastasis	Ventricular filling becomes slow.	AV valves remain open.
Last rapid filling	Atrial systole pushes final blood into ventricles.	AV valves open.

Applied note

Heart sounds and murmurs are understood by knowing valve closure and opening during different phases of the cardiac cycle.

Important cardiac values

Value	Approximate normal value	Meaning
End-diastolic volume (EDV)	About 120-130 mL	Blood in ventricle at end of diastole.
End-systolic volume (ESV)	About 50-60 mL	Blood remaining after systole.
Stroke volume	About 70 mL	Blood pumped by each ventricle per beat.
Ejection fraction	About 55-70%	Percentage of EDV ejected during systole.
Cardiac output	About 5 L/min	Heart rate x stroke volume.

2(g). Normal ECG

Definition

Electrocardiogram (ECG) is the graphic recording of electrical activity of the heart. It is useful for assessing rhythm, conduction and myocardial abnormalities.

Waves, segments and intervals

ECG part	Represents	Normal value / point
P wave	Atrial depolarization	Duration about 0.1 sec; amplitude about 0.1-0.12 mV.
QRS complex	Ventricular depolarization	Duration about 0.08-0.10 sec.
T wave	Ventricular repolarization	Duration about 0.2 sec; amplitude about 0.3 mV.
PR interval	Atrial depolarization + AV nodal conduction	About 0.12-0.20 sec.
ST segment	Isoelectric period after ventricular depolarization	Normally at baseline; elevation/depression is clinically important.
QT interval	Ventricular depolarization + repolarization	About 0.4-0.42 sec.
RR interval	Time between two R waves	Used to calculate heart rate.

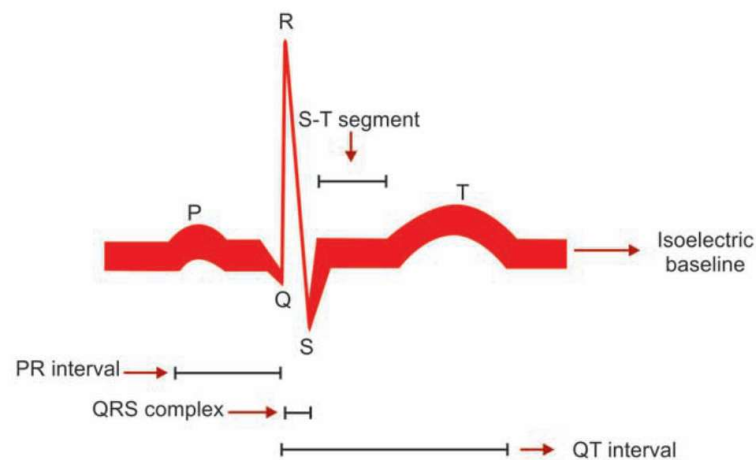
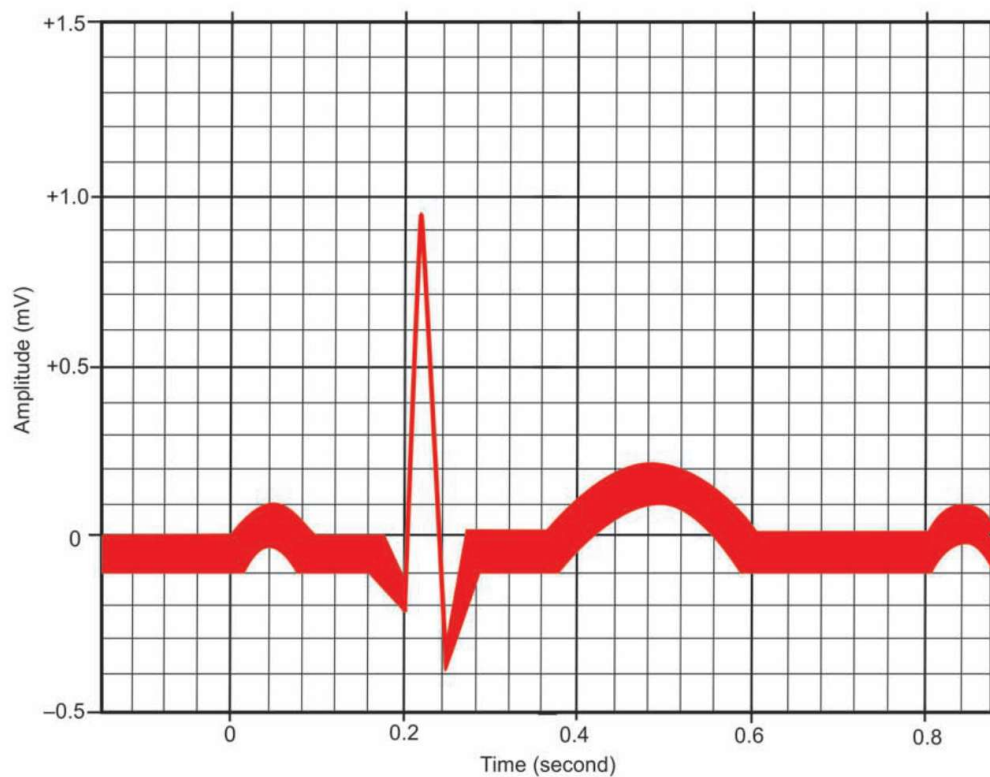


FIGURE 94.3: Waves of normal ECG

Source diagram: Waves, intervals and segments of normal ECG (Sembulingam, Fig. 94.3).

Clinical importance

- Detects arrhythmias and heart block.
- Helps identify myocardial infarction and ischemia.
- Shows chamber enlargement and electrolyte disturbances.
- Used to monitor heart rate and rhythm during emergencies.

ECG exam tips

Tip	Use
Heart rate from ECG	Heart rate = $1500 / \text{number of small squares between two R waves}$, or $300 / \text{large squares}$.
PR interval prolonged	Suggests AV nodal conduction delay or heart block.

Tip	Use
ST elevation	May indicate acute myocardial infarction when present in appropriate leads.
Tall T wave	May occur in hyperkalemia.
Absent P wave with irregular rhythm	Suggests atrial fibrillation.

Quick revision checklist

Topic	Must-remember line
O ₂ -Hb curve	Sigmoid curve; right shift means easier O ₂ unloading to tissues.
Cell membrane	Fluid mosaic model: lipid bilayer with embedded proteins.
WBC	Defence depends on diapedesis, chemotaxis, phagocytosis and immunity.
Acquired immunity	Specific immunity with memory; T cells = cellular, B cells = humoral.
Saliva	Moistening, taste, starch digestion, oral cleansing and antibacterial action.
Cardiac cycle	0.8 sec at 72/min; S ₁ = AV valve closure, S ₂ = semilunar valve closure.
Normal ECG	P = atrial depolarization, QRS = ventricular depolarization, T = ventricular repolarization.

Last-minute exam tip

For short notes, draw a simple labelled diagram wherever possible. Diagrams for O₂-Hb curve, cardiac cycle and ECG are especially scoring.

Likely one-line answers and MCQ traps

Question / clue	Answer to remember
Example of active transport	Sodium-potassium pump (Na ⁺ -K ⁺ ATPase).
Pepsinogen is secreted by	Chief cells/peptic cells of gastric glands.
Genetically programmed cell death	Apoptosis.
Gastrin is secreted by	G cells of pyloric antrum.
First heart sound	Closure of AV valves: mitral and tricuspid.
Second heart sound	Closure of semilunar valves: aortic and pulmonary.
Surfactant is secreted by	Type II alveolar epithelial cells.
Voluntary phase of swallowing	Oral phase.
Involuntary swallowing phases	Pharyngeal and oesophageal phases.
Haemophilia B is due to deficiency of	Factor IX.
Milieu interior was coined by	Claude Bernard.
Phagocytosis sequence	Chemotaxis -> diapedesis -> opsonization -> phagocytosis.

How to score better in short notes

- Start with a clean definition; it immediately gives structure to the answer.
- Use tables for comparisons such as active/passive immunity and right/left shift.
- Mention one applied point at the end: e.g., O₂-Hb curve in exercise, ECG in myocardial infarction, saliva in xerostomia.
- Underline key terms in your copy: P₅₀, Bohr effect, glycocalyx, diapedesis, memory cells, S₁/S₂, QRS complex.

Diagram checklist for fast revision

Topic	Draw / label these parts
O ₂ -Hb curve	Axes: PO ₂ and Hb saturation; mark right shift, left shift and P ₅₀ .

Topic	Draw / label these parts
Cell membrane	Phospholipid bilayer, cholesterol, integral protein, peripheral protein and glycocalyx.
WBC	Flow: chemotaxis -> diapedesis -> opsonization -> phagocytosis.
Acquired immunity	Antigen -> B/T lymphocytes -> plasma/cytotoxic cells -> memory cells.
Saliva	Major glands: parotid, submandibular and sublingual; add enzymes and mucus.
Cardiac cycle	Show systole, diastole, valve closure and heart sounds S1/S2.
ECG	Label P wave, QRS complex, T wave, PR interval, ST segment and QT interval.

Common mistakes to avoid

- Do not write right shift as increased affinity; right shift means decreased Hb-O₂ affinity and better unloading.
- Do not confuse humoral immunity with T cells; humoral immunity is mainly B cells and antibodies.
- Do not write first heart sound as semilunar valve closure; S1 is AV valve closure.
- Do not write T wave as depolarization; T wave is ventricular repolarization.
- For saliva, mention both digestive and protective functions; writing only digestion makes the answer weak.

Mini memory aids

- P-QRS-T = atria depolarize, ventricles depolarize, ventricles repolarize.
- S1 = AV valves shut; S2 = semilunar valves shut.
- B cells = antibodies in Blood/body fluids; T cells = direct Tissue/cell attack.
- Right shift = oxygen Released to tissues; Left shift = oxygen Locked with Hb.
- Neutrophils fight bacteria first; eosinophils fight parasites and allergy.